

Calculus 2023 FRQ

fast sloth

May 2026

1 Question 1

1.1 A

$\int_{60}^{135} f(t)dt$ is the amount of gallons of gasoline pumped into the gas tank from 60 seconds since pumping began to 135 seconds since pumping began

$$\int_{60}^{135} f(t)dt \approx 30(0.15) + 30(0.1) + 15(0.05)$$

1.2 B

According to the Mean Value Theorem, because $f(t)$ is differentiable on the interval $(60, 120)$ and $f(60)$ and $f(120)$ are defined, implying continuity on the interval $[60, 120]$ then $\exists c \in (60, 120) : f'(c) = \frac{0.1-0.1}{60-120} = 0$

1.3 C

The average flow rate of gasoline over the time interval $0 \leq t \leq 150$ can be defined as $\frac{g(150)-g(0)}{150-0} = 0.0000165924632477$

1.4 D

$g'(140) = -0.00623319244724$ gallons per second squared is the instantaneous rate of change of the flow rate at time $t = 140$ seconds

2 Question 2

2.1 A

$$\begin{aligned}y(t) &= 2 \sin(t) \rightarrow y'(t) = 2 \cos(t) \rightarrow y''(t) = -2 \sin(t) \\x'(t) &= e^{\cos(t)} \rightarrow x''(t) = -\sin(t)e^{\cos(t)} \\a(t)|_{t=1} &= (-2 \sin(1), -\sin(1)e^{\cos(1)})\end{aligned}$$

2.2 B

$$\text{Speed} = \sqrt{x'(t)^2 + y'(t)^2} = 1.5 \rightarrow t = 1.25447$$

2.3 C

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}}|_{t=1} = 0.629530109581 \text{ The x position of time at time } t = 1 \text{ is } = \\1 + \int_0^1 \frac{dx}{dt} &= 3.34157484171\end{aligned}$$

2.4 D

$$\text{Total distance } S = \int_0^\pi \sqrt{x'(t)^2 + y'(t)^2} dt = 6.03461133685$$

3 Question 3

3.1 A